

# JIAO GUO

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## EXPERIENCE

### **Geospatial Research Coordinator**, University of Calgary & ABMI

September 2022 – August 2024 (full-time); September 2024 – June 2025 (part-time)

- Coordinating annual research workshop, summer fieldtrips (preparation, safety check, wrap-up) of Boreal Ecosystem Recovery and Assessment (BERA) program.
- Assisting remotely sensed data acquisition (PlanetScope orders), distribution (ABMI's LiDAR) and management (LiDAR processing in R/LasTools, satellite images co-registration in Python) between Alberta Biodiversity Monitoring Institute (ABMI) and BERA
- As a member of ABMI's Earth-Observation (EO) Insight team, assisting on updating annual spectral regeneration product of forest Harvest Areas of ABMI's human footprint Inventory (HFI) for 2020 and 2021; testing performance of PlanetScope images of algal bloom monitoring in Alberta lakes.
- Coordinating ABMI and BERA's monthly social events; facilitating regular technical interactions between BERA and ABMI staff; exchanging data between ABMI and BERA group members.
- Seeking and coordinating external research partners (e.g., Alberta Energy Regulator) and funding application process led by Applied Geospatial Research Group (AGRG) and ABMI.

### **Associate Machine Learning Developer, AltaML**

September 2024 – current (Contract)

- Labelling high-resolution and hyperspectral satellite images using object-based classification workflow for feeding into machine learning models.
- Geometric and radiometric correction of remotely sensed images

### **Remote Sensing Analyst**, NWT Centre for Geomatics, Government of Northwest Territories

August 2020 – August 2022 (full-time)

- Satellite & airborne-based image processing and distribution to Government clients.
- Spatial data management include storing, processing, publishing, server testing, and metadata generation. Major author of raster data management guide for the centre.
- Assisting long-term landscape change detection using cloud-based Landsat stacks; helping intergovernmental and academia to identify the changes using Landsat derived product.
- Spatial analysis for projects including landfill airspace capacity, hiking trail cost-distance simulations, object-based classification for glaciers and ice patches, flood inundation mapping.
- Operational mapping for wildfire control during the summer seasons.

### **Research Assistant**, Canadian Mountain Network

July 2020 – Current (part-time)

- Task-based mapping and spatial analysis for Canadian mountain environment, working with Dr. Graham McDowell, who led the Canadian Mountain Assessment (CMA) initiative.
- GIS analyst and cartographer for *Hills Thought to be Mountains* project led by Canadian Mountain Network's co-research director, Dr. Murray Humphries at McGill University.

### **GIS Specialist; Research Assistant**, Department of Geography, University of Calgary

December 2018 – August 2020

- Developed / implement a mapping "toolkit" for the Accelerating Implementation of RenewableEnergy (AI-RE) project in Canmore, Alberta, funded by Community Energy Knowledge-Action Partnership (CEKAP).
- Using spatial analysis to understand sidewalk-width patterns and generate a web-mapping app for promoting a walkable & safe Calgary during the COVID-19 pandemic.
- Developed and still maintain map series of influenza outbreaks in Alberta for TARRANT Viral Watch team, Family Medicine at Cumming School of Medicine.

## SKILLS

**GIS:** ArcGIS Desktop, ArcGIS Online, ArcGIS Pro, FME, QGIS, Whitebox Geospatial  
**Programming/Web language:** Python ([sample work](#)), JavaScript, HTML, Markdown  
**Remote sensing:** PCI Geomatics, Erdas Imagine, ENVI, eCognition Developer, LASTools, CloudCompare, Google Earth Engine API, ESA SNAP Tool, DNRGarmin  
**Database management:** MySQL, PostgreSQL/PostGIS  
**Statistical tool:** R, Python

## PUBLICATION

- Guo J, Fast V, Teri P, Calvert K.** (2020) Integrating Land-Use and Renewable Energy Planning Decisions: A Technical Mapping Guide for Local Government. *ISPRS International Journal of Geo-Information*, 9(5):324. <https://doi.org/10.3390/ijgi9050324>
- Guo, J., & Fast, V.** (2019). Growing Up, Growing Out: Comparing Spatial Patterns of Urban Populations in Canada. *Geomatica*, 73(2): 45-61. <https://doi.org/10.1139/geomat-2019-0011>
- Guo, J., & Fast, V.** (2019). Spatial Insights on Urban Density: A Case Study of Calgary. Proceedings of the 7th Conference on Spatial Knowledge and Information Canada, 7(3), 2. <http://ceur-ws.org/Vol-2323/>
- Fast, V., & Guo, J.** (2021). Putting Pedestrians First: Sidewalk Infrastructures, Width Patterns and COVID-19. *GI Forum*, 9(2):242-250. [https://doi.org/10.1553/giscience2021\\_02\\_s242](https://doi.org/10.1553/giscience2021_02_s242)
- McDowell, G., & Guo, J.** (2021). A Nationally Coherent Characterization and Quantification of Mountain Systems in Canada. *Mountain Research and Development*, 41(2), R21. <https://doi.org/10.1659/MRD-JOURNAL-D-20-00071.1>
- McDowell, G., Stevens, M., Marshall, S., Higgs, E., Jacob, A., Johnson, G. M. J., ... & Lancaster, S. (2023).** Canadian Mountain Assessment: Walking Together to Enhance Understanding of Mountains in Canada. In *Canadian Mountain Assessment*. University of Calgary Press. <https://ucp.manifoldapp.org/projects/9781773855103> (Chapter contributor)
- Humphries, M. M., Bowser, A. K., Guo, J., & Menzies, A. K. (2024).** Hills thought to be mountains: A geobiocultural characterization of island highlands in Canada's continental plain. *The Canadian Geographer/Le Géographe canadien*, 69(1), e12964. <https://doi.org/10.1111/cag.12964>
- Carpino, O., Wright, S., Guo, J., van der Sluijs, J., Connon, R., & Quinton, W. (2025).** Detecting Forest-Wetland Changes in Northwestern Canada's Discontinuous Permafrost Region Using Landsat Trend Analysis. *Ecohydrology*, 18(1), e2759. <https://doi.org/10.1002/eco.2759>

## EDUCATION

**Doctor of Philosophy**, University of Calgary, Calgary AB

September 2024 – current

- Research on peatland disturbance from mineral roads using drone-based remote sensing

**Master of Geographic Information Systems**, University of Calgary, Calgary AB

September 2017 – August 2019

- Courses taken include advanced GIS, GIS-based Python programming, urban GIS, advanced remote sensing, spatial-statistics. Thesis: "Mapping Renewable Energy Capacity in Canmore, Alberta: Implementing a Standardized Mapping Framework and GIS-based MCDA".

**Bachelor of Science, Environmental Science**, University of Western Ontario, London ON

September 2013 – April 2017

- Courses taken include GIS, decision making in GIS, cartography, and remote sensing. Honor thesis: "Estimating the Complete Surface Temperature of the Greater Toronto Area" using MODIS thermal bands and Environment Canada's Surface Prediction System.

## CONFERENCES & PUBLIC SPEECH

### **ArcticNet's Annual Scientific Meeting (ASM)**

December 2025 – Calgary, AB

- Poster Title: Where's North in Canada?

### **Alberta Society of Professional Biologists (ASPB) Conference**

November 2025 – Banff, AB

- Poster Title: Edge Effects of mineral roads on woody plant communities in northern Alberta: Insights from high-resolution remote sensing

### **Canada's National Geomatics Expo 2024**

October 2024 – Calgary, AB

- Title: Democratizing and Operationalizing Satellite-based Vegetation Recovery Monitoring in Canadian Boreal Forests

### **Aurora Research Institute Summer Speaker Series**

August 2022 – Inuvik, NT

- Title: Long-Term Change Detection using Landsat Satellites

### **GI-Forum 2021**

August 2021 – Vienna, Austria (Virtual)

- Title: Putting Pedestrians First: Sidewalk Infrastructures, Width Patterns, and COVID-19

### **56<sup>th</sup> Annual Geography Department Conference (University of Calgary)**

March 2019 – Calgary, AB

- Title: Comparative urban density: analyzing Calgary's 14 new communities

### **7th Conference on Spatial Knowledge and Information Canada**

February 2019 – Banff, AB

- Title: Spatial Insights on Urban Density: A Case Study of Calgary  
Best Presentation Award at Spatial Knowledge and Information (SKI) Canada

### **Canadian Association of Geographers, Ontario Divisions (CAGONT) 2019**

November 2019 – Guelph, ON

- Title: Mapping Renewable Energy Opportunities for a Small Municipality: A Case Study in Canmore, AB

### **Canada Energy Regulator: GIS Day Conference, Calgary (2019)**

November 2019– Calgary, AB

- Title: Land-based Renewable Energy Mapping: A Local Planning Guide for Canmore, Alberta

## SCOLARSHIP

**Alberta Graduate Excellence Scholarship** (Fall 2024, \$11,000)

**Faculty of Graduate Studies Doctoral Scholarship** (Fall 2025, \$15,000)

**NSERC CGS-Doctoral Award** (Fall 2026, \$40,000 annually)

## EXTRACURRICULAR ACTIVITIES

**Firefighter (on-call volunteer-based)**, Fire Department, Town of Inuvik

September 2021– August 2022

- Have 300+ hours of standard firefighter training; responded to fire calls as paged by the chief officer; fire alarms and extinguishers and helped fire prevention awareness education to the local communities.

**Weekly Volunteer**, Wing Kei Senior Care Centre (Calgary)

October 2018 –March 2020 (suspended due to COVID-19)

- Assist the recreational programs to accompany elderly residents (300+ hours).

**Weekly Volunteer**, University of Calgary

September 2018 – April 2019

- Student Wellness Centre: International gradstudents' peer support.
- International Student Services: guides and provides resources to international students.